Alcohol Metabolism

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Objectives:

- At the end of this lecture, it is expected to;
  - State the Biomedical Importance of alcohol metabolism
  - Describe and write the reactions of alcohol metabolism
  - Explain the relations between alcohol metabolism and gluconeogenesis
  - Explain the relations between alcohol metabolism and FA oxidation
  - Explain the effects of alcohol on liver metabolism

BioMedical Importance of Alcohol Metabolism

1. It serves as energy source like carbohydrate.
   - However, if consumed excess (chronic alcoholism) leads to fatty liver development.
2. Some individuals (chronic alcoholics) develop vit A and thiamin deficiency.
3. Lactic acidemia occurs due to excess NADH in the cytosol.
4. Moderate consumption of alcohol was found to be beneficial to stroke patients.
   - HDL level is higher in people consuming small amount of alcohol daily.
5. Clearance of toxins from circulation by liver becomes slow
   - Some compounds get converted to carcinogens in chronic alcoholics.
6. Alcohol consumption aggravates gout and porphyrias.

ALCOHOL METABOLISM

Major pathway:

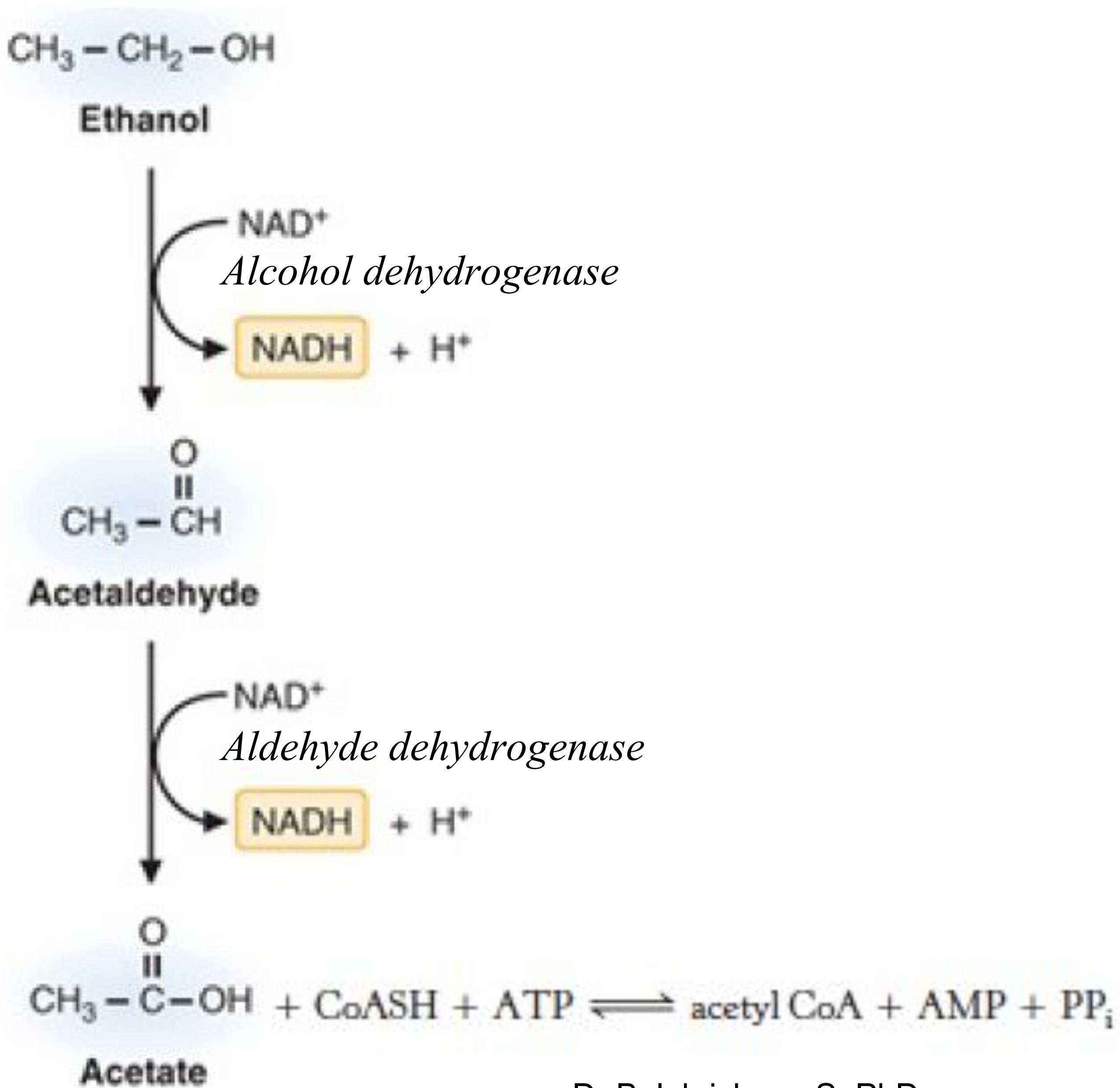
- Alcohol cannot be excreted and must be metabolized, primarily by the liver.
- This metabolism comprises two steps:
  - The first step, catalyzed by the enzyme *alcohol dehydrogenase*, takes place in the cytoplasm
  - The second step, catalyzed by *aldehyde dehydrogenase*, takes place in mitochondria.
- Acetate, once produced, is not toxic, so a slow-acting acetyl-CoA synthetase (which uses acetate as a substrate) converts into acetyl-CoA .

$$\text{Acetate} + \text{CoASH} + \text{ATP} \rightleftharpoons \text{acetyl CoA} + \text{AMP} + \text{PP}_i$$

ALCOHOL METABOLISM - MAJOR PATHWAY

Cytoplasm

Mitochondria



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ALCOHOL METABOLISM (CONT'D)

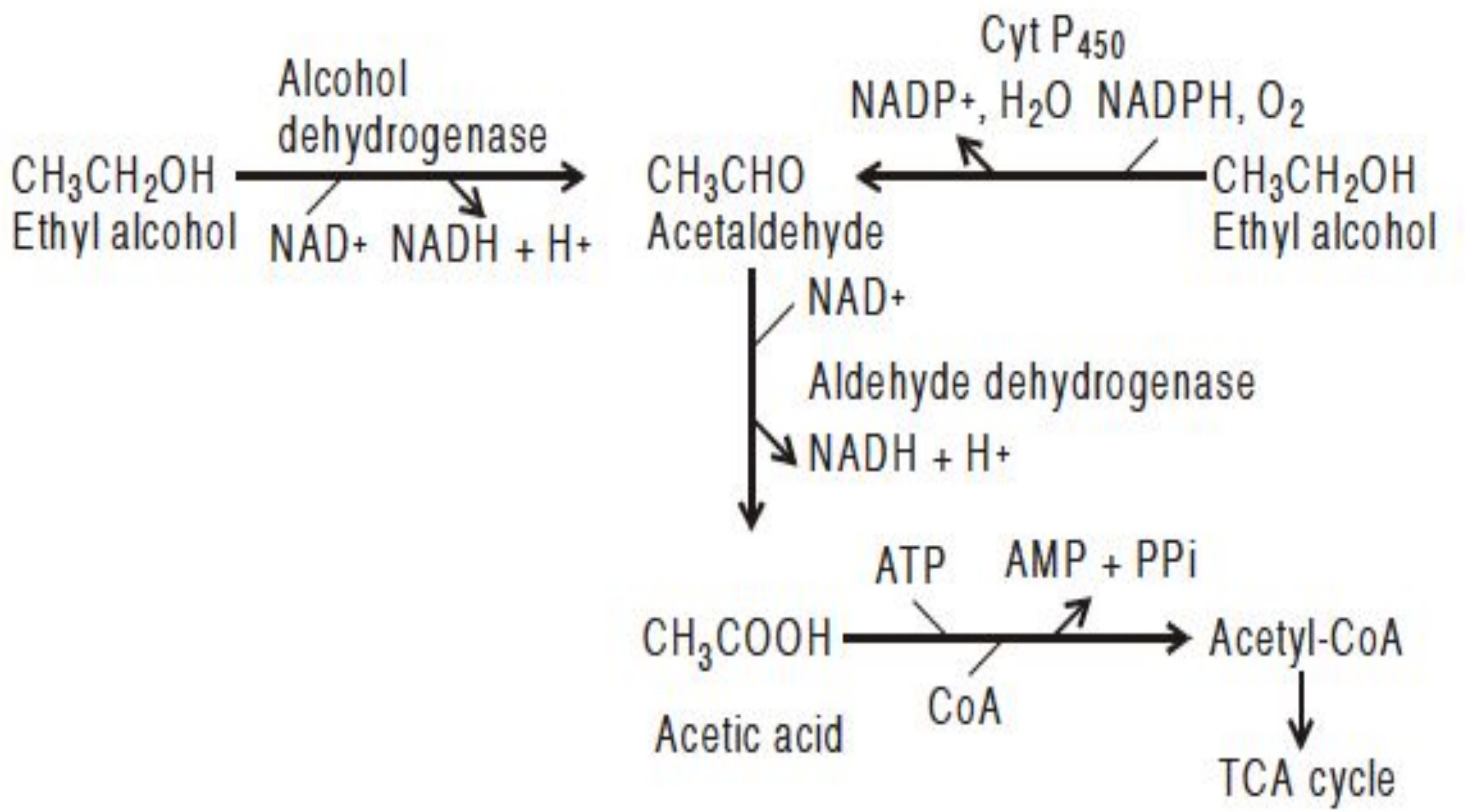
Minor pathway:

- Some alcohol is metabolized by the cytochrome P-450 system:

$$H_3C\text{-}CH_2OH + O_2 + NADPH + H^+ \,(\text{Ethanol})$$
$$\downarrow$$
$$H_3C\text{-}CHO + NADP^+ + 2H_2O\,(\text{Acetaldehyde})$$

- Most people metabolize about 10 g of alcohol per hour, and the blood alcohol level decreases by about 0.15 g/L every hour.
  - These calculations are important when a blood sample from a drunken driver has been obtained some hours after an accident.

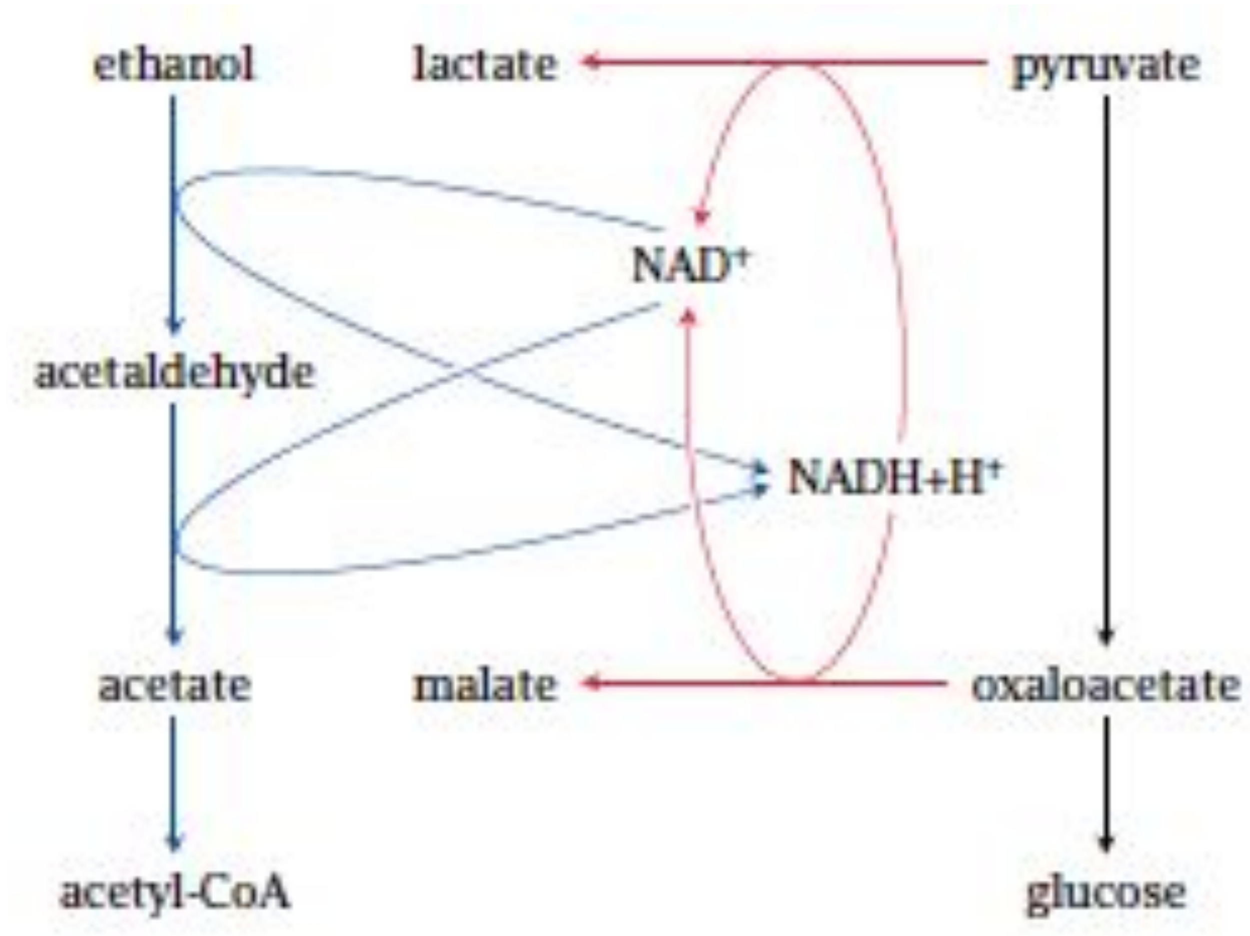
ALCOHOL METABOLISM



Alcohol metabolism Vs Gluconeogenesis

- Note that Alcohol consumption leads to an accumulation of NADH.
- This **high conc. of NADH inhibits Gluconeogenesis**
- In fact, high conc. of NADH will cause the reverse reaction to predominate.
- High NADH favors the formation of:
  - Lactate from pyruvate
  - Malate from OAA in the cytoplasm
  - Glycero13-phosphate from DHAP
- The consequences may be **hypoglycemia and lactic acidosis.**

Ethanol degradation inhibits gluconeogenesis



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<u>Alcohol metabolism Vs Fatty acids (FA) oxidation</u>

- The **NADH glut also inhibits FA oxidation.**
- The **purpose of FA oxidation is to generate NADH for ATP** by oxidative phosphorylation, but an alcohol consumer's NADH needs are met by ethanol metabolism.
- In fact, the excess NADH signals for FA synthesis. Hence, TG accumulates in liver, may lead to **Fatty liver.**

Effects of alcohol on liver metabolism

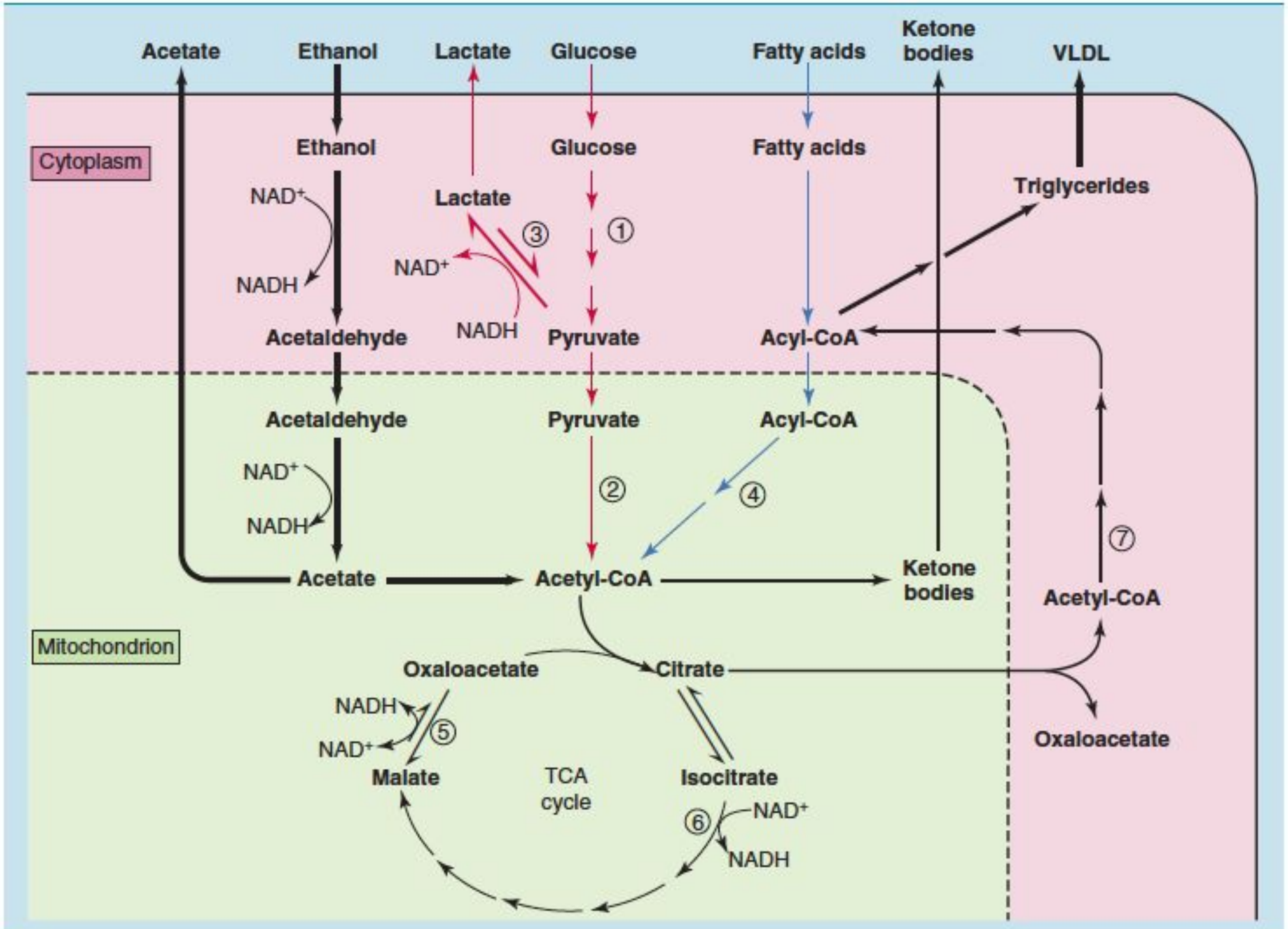

Fig. 32.24 Effects of alcohol on liver metabolism. Important control points: (1) phosphofructokinase: inhibited by high energy charge; (2) pyruvate dehydrogenase: inhibited by high energy charge, high NADH/NAD$^+$ ratio, and high acetyl-coenzyme A (*CoA*); (3) lactate dehydrogenase: high NADH/NAD$^+$ ratio favors lactate formation; (4) β-oxidation: slowed down because of low NAD$^+$; (5) malate dehydrogenase: high NADH/NAD$^+$ ratio favors malate formation, oxaloacetate is depleted; (6) isocitrate dehydrogenase: inhibited by high energy charge and high NADH/NAD$^+$ ratio; and (7) acetyl-CoA carboxylase: may be stimulated by high citrate (accumulates because isocitrate dehydrogenase is inhibited). *TCA*, Tricarboxylic acid; *VLDL*, very-low-density lipoprotein.

*Principles of medical biochemistry / Gerhard Meisenberg, William H. Simmons.

Read more…

THANK YOU…